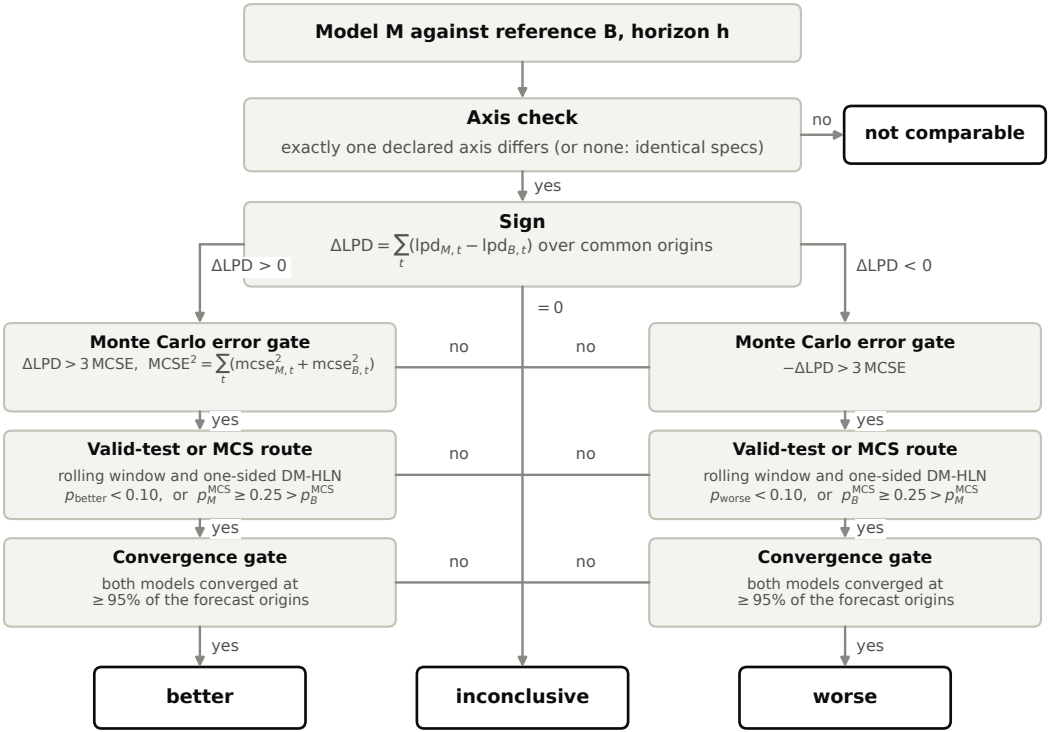
Model M against reference B, horizon h
Axis check
exactly one declared axis differs (or none: identical specs)
no
not comparable
yes
Sign
$\Delta\mathrm{LPD} = \sum_t (\mathrm{lpd}_{M,t} - \mathrm{lpd}_{B,t})$ over common origins
$\Delta\mathrm{LPD} > 0$
$= 0$
$\Delta\mathrm{LPD} < 0$
Monte Carlo error gate
$\Delta\mathrm{LPD} > 3\,\mathrm{MCSE}$, $\mathrm{MCSE}^2 = \sum_t (\mathrm{mcse}^2_{M,t} + \mathrm{mcse}^2_{B,t})$
Monte Carlo error gate
$-\Delta\mathrm{LPD} > 3\,\mathrm{MCSE}$
no
no
yes
yes
Valid-test or MCS route
rolling window and one-sided DM-HLN
$p_{\text{better}} < 0.10$, or $p_M^{\mathrm{MCS}} \geq 0.25 > p_B^{\mathrm{MCS}}$
Valid-test or MCS route
rolling window and one-sided DM-HLN
$p_{\text{worse}} < 0.10$, or $p_B^{\mathrm{MCS}} \geq 0.25 > p_M^{\mathrm{MCS}}$
no
no
yes
yes
Convergence gate
both models converged at
≥ 95% of the forecast origins
Convergence gate
both models converged at
≥ 95% of the forecast origins
no
no
yes
yes
better
inconclusive
worse